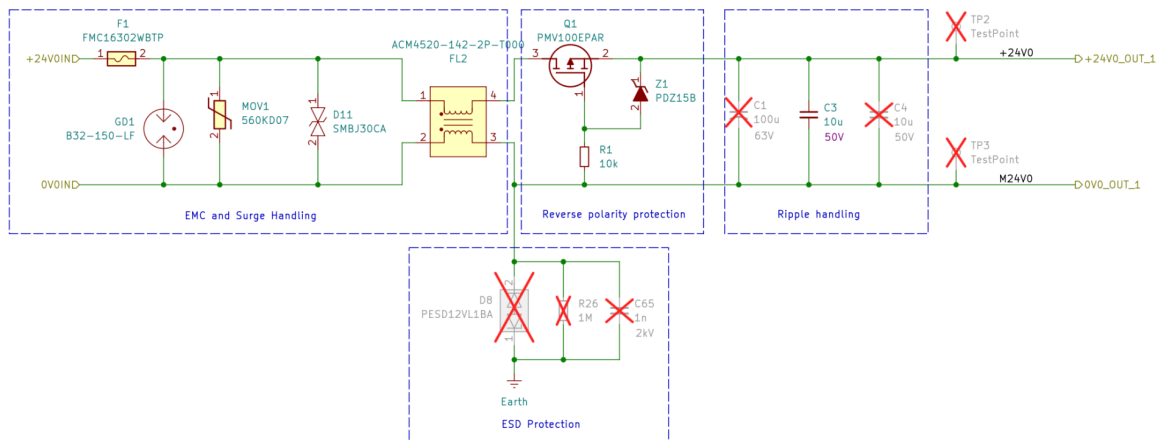




Recommended improvement: Add safety-rated **Y capacitors (bypass capacitors)** from both **DC lines to Protective Earth (PE)/chassis** at the **entry and exit points of the common-mode choke**.

- At the **input side of FL2**, provide:
 - +24 V → Y capacitor → PE/Chassis
 - 0 V → Y capacitor → PE/Chassis
- At the **output side of FL2**, provide:
 - +24 V → Y capacitor → PE/Chassis
 - 0 V → Y capacitor → PE/Chassis

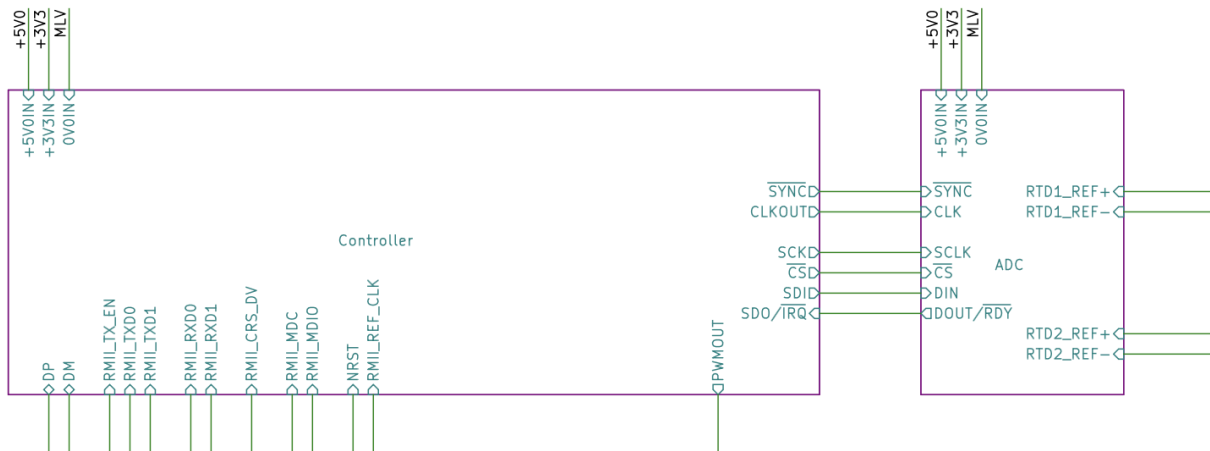
These capacitors provide a low-impedance path for high-frequency common-mode noise to chassis/earth while maintaining DC isolation from PE. They can significantly improve conducted EMI performance.

Note:

Both side placement is not always required: The purpose of the specific block is not to filter out random noise but, to provide GND path to high voltage discharge.

Ref:

<https://learnemc.com/emc-blog-02>
TE04EA-1 (Basics of EMI Filters by Murata)



Circuit Comment:

The CPU interface to the 24-bit ADC should preferably be implemented through a **galvanically isolated digital interface**. Since the 24-bit ADC is used to acquire very low-level and highly sensitive analog signals, digital switching noise generated by the CPU, SPI interface, clock, and other digital circuits can couple into the ADC and degrade measurement accuracy and resolution.

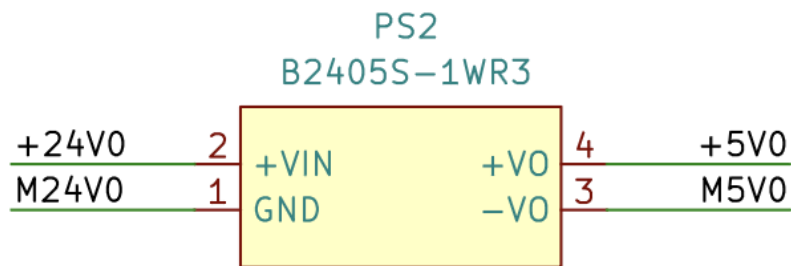
Therefore, **galvanic isolation should be provided between the CPU/digital ground and the ADC/analog ground** to prevent digital noise, ground-loop currents, and common-mode interference from affecting the sensitive ADC measurements. This will help maintain the required signal integrity and maximize the effective resolution of the 24-bit ADC.

Note:

Split ground is not found recommended in any doc from the ADC manufacturer.

Ref.:

Datasheet of AD7124-8
Vol 56, No 3—September 2022 by Analog Devices
MT-031 by Analog Devices



The proposed 24 V DC to 5 V DC converter is a switching-type power converter and may introduce significant switching noise, ripple, and high-frequency EMI into the 5 V supply. Such noise can adversely affect the performance and measurement accuracy of the **24-bit ADC**, particularly when measuring very low-level and sensitive signals.

Note:

24V DC to 5V DC supply is the first stage buck, all the digital and analog chips runs on 3.3V DC which is derived from the 5V supply. Ferrite beads and capacitors have been used for filtering of the ripples.